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Climate Change and Food Insecurities: The Importance of Food Loss and Waste Reduction in Indonesia

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Abstract. Rapid increasing population of Indonesia poses a future threat to the food supply for the national demand. Furthermore, global climate change is putting a strain on the food supply chain. Droughts, floods, landslides, and pest/disease attacks all have serious consequences for agricultural production, particularly food crops. According to The Economist Intelligence Unit (EIU) data published in 2017, Indonesia ranks second in the world in terms of food loss and waste contribution, chasing only Saudi Arabia. Moreover, according to the Food Sustainability Index in the Economist Intelligence Unit (2017), Indonesian food loss and waste or uneaten totalled 1.3 million tons in a single year. In Indonesia, the most common method of dealing with food waste is dumping or landfilling, which produces methane and other greenhouse gases that contribute to climate change. This research was conducted by employing quick survey for colecting primary data and desk evaluation study for collecting secondary data. Meta analysis was employed in this research. Therefore, to emphasize the importance of reducing food loss and waste in Indonesia to reduce food insecurity caused by climate change, this paper investigates how food loss and waste exacerbates climate change. This paper outlined some scenarios for preventing food loss and waste at all stages of the food chain, including agricultural production, post-harvest handling and storage, distribution and consumption behaviour, and government policy, which can be implemented to begin initiatives to reduce food loss and waste to address the threat of future starvation in Indonesia because of climate change.

1. Introduction

Indonesia, as a developing country, is still dealing with food issues, especially the issue of supplying food due to rapid population growth, which increases the demand for supplies. The total population of Indonesia is currently 277.26 million people, with an annual increase of 1.2, ranking Indonesia fourth in the world in terms of population [1]. Besides that, Indonesia's population is estimated to increase 330.90 million by 2050, implying that food demand will rise even further. Expanding agricultural productivity is thus one method of achieving food security and avoiding a food insecurity affected by increased population. Global climate, as well as non-climate stressors (e.g., population and income growth, demand for animal-sourced products), are putting strain on the food supply chain [2]. Droughts, floods, landslides, and pest/disease attacks caused by climate change have severe consequences for agricultural production, especially food crops. As a result, future demand can be met with a smaller increase in agricultural production by making better use of the food that is currently available at the current level of production.

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Nonetheless, facing difficult efforts to increase food production in Indonesia have not been accompanied by initiatives to reduce food loss and waste. Food availability or security efforts commonly concentrate solely on how to maximize agricultural production while disregarding how to prevent food loss and waste [3] . This tendency can be seen in initiatives to improve food supply, which are still focused on increased production, whether through increased agricultural output or expanded planted areas. Meanwhile, efforts to reduce food waste have been poorly managed. On the other hand, discovered that the current global farm of crop production, approximately at 4600 kcal per person each day [4], should be enough to feed the current population and more. Food waste, both post-harvest and consumer-generated, is expected to cost 1400 kcal per person, or one-third of all food produced for human consumption [5]. Even so, according to data published in 2017 by The Economist Intelligence Unit (EIU), Indonesia ranks second in the world regarding food loss and waste contribution, trailing only Saudi Arabia. Moreover, according to the report from the Food Sustainability Index Food Sustainability Index in the Economist Intelligence Unit (2017), Indonesian food waste or uneaten totalled 1.3 million tons in one year. As a result, the average person dumps 300 kg of food each year. Given these barriers, as well as growing concerns about feeding more than 300 million Indonesians by 2050, climate change, water shortages, and resource scarcity, addressing the problem of food waste ought to be a primary concern for various organizations and institutions, as well as policymakers.

In response to the importance of reducing food loss and waste in Indonesia to reduce food insecurity caused by climate change, this paper examines how food loss and waste exacerbates climate change. Then it identifies food loss and waste data in Indonesia and around the world. Data and information are collected from research results, journals, online databases, as well as related publications and descriptively analysed to ensure their relevance to the purpose of this paper. Following that, the significance of incorporating minimizing food loss and waste is emphasized, with the findings, recommending several strategies that may be implemented to begin the initiatives to minimize food loss and waste to tackle the threat of future starvation in Indonesia caused by climate change.

2. Research Methodology

The primary data used in this study were collected from a rapid survey of several agricultural production centers in Bantaeng Regency, South Sulawesi. The secondary data used in this study were obtained from desk evaluation results from various government and non-government institutions. Data processing is done quantitatively and using meta-analysis.

3. Food loss and waste generation in Indonesia

Food waste or food loss is a worldwide issue that has existed for a long time but has recently become more troublesome because it reflects one of the most critical social, economic, and ecological challenges confronting humanity, as well as an ethical issue [6]. Food loss is defined by the Food and Agriculture Organization of the United Nations [7] as the transfer of a specific amount of food from the producer supply chain to the market (food chain). Food loss can be caused by issues with the pre-harvest process, storage, processing, and packing of the food, and producers who decide to dump the food because the market rejects it.

Food loss occurs in developing countries, including Indonesia, due to a high level of agricultural production that is not adjusted with adequate technology, resulting in food that is damaged in the process or is not of the quality desired by the market before it reaches the hands of consumers. Food waste, on the other hand, is food that is ready for human consumption but is discarded with no apparent reason or just because it is about to expire.

Despite frequent warnings not to leave food unattended and throw it away, Indonesia holds the second position in related to food waste, one rank below Saudi Arabia. Food waste in Indonesia totals 1.3 million tons per year, or 300 kilograms per person, according to Central Statistics Agency (BPS) data from 2015. When converted to rupiah, every 1.3 million tons of food waste generates IDR 27 trillion. Nonetheless, 19.6 million Indonesians remain undernourished. If consumed, that 27 trillion could feed 28 million people, or 11% of the total population over a year. Obviously, it may have the

possibility to be a solution to stunting and malnutrition problem in Indonesia, which impacts 30% of the population.

Food loss and waste, ironically, are the leading sources of waste in Indonesia. Food waste accounts for 39.74% of total waste in Indonesia, according to data from the National Waste Management Information System of Indonesia (SIPSN) in 2020 (see Figure 1). When food waste is broken down by source (see Figure 2), households are the most significant contributors.

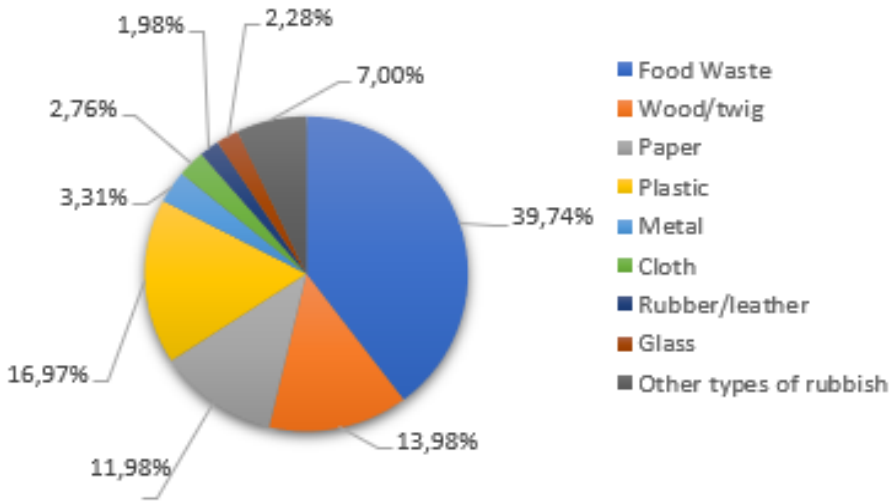

Figure 1. Waste distribution (%) in Indonesia based on its type in 2020 [8]

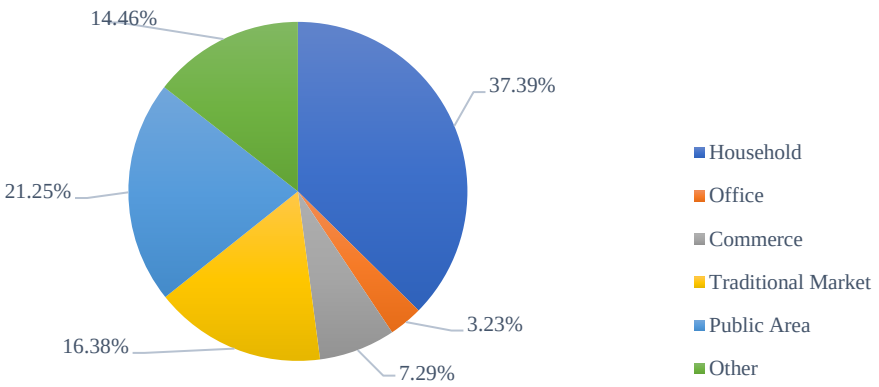

Figure 2. Waste distribution (%) in Indonesia based on its source in 2020 [8]

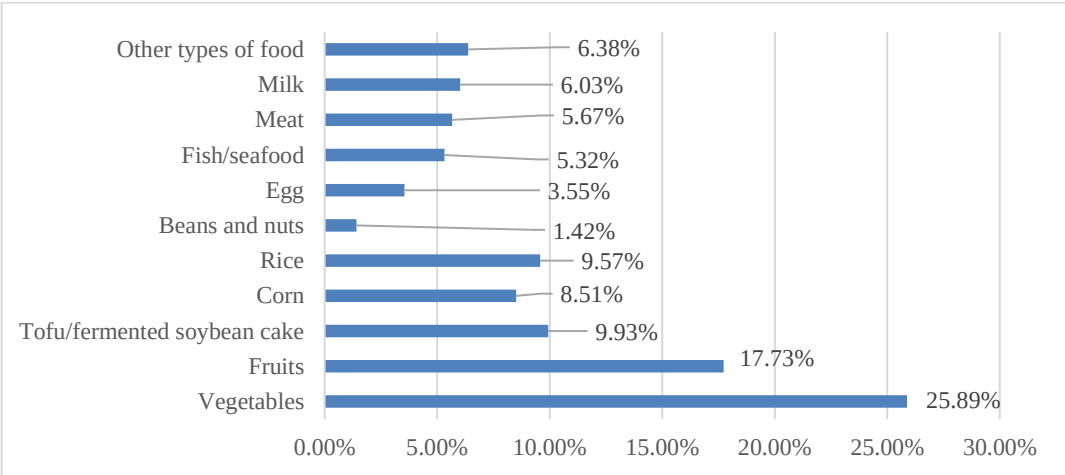

Figure 3. Estimated commodity types of food waste distribution per person per year [9]

According to the picture above, households generate up to 113 kilograms of food waste per year out of a total of 1.3 million tons of food waste. According to figure 3, each Indonesian dumps an average of 28 kg of food per year, with vegetables, accounting for 25.89%, fruits accounting for 17.73%, and tofu/fermented soybean cake accounting for 9.93%. In Indonesia, it appears that vegetables are a major source of waste. This is mainly due to waste generated during agricultural production, post-harvest handling and storage, and consumer consumption. Indonesia's significant contribution stems from its dominance in global vegetable production and consumption, responsible for more than half of both [5]. This is a particularly sensitive issue in Indonesia, where millions of people remain malnourished and impoverished. Poorness affects 11% of the population, and malnutrition affects 7.6% of the 260 million people. Only Ethiopia (32 percent) and India outperform Indonesia (15.2%) in this category [7].

4. The impact of food loss and waste on climate change

Food loss and waste have a huge impact on society, economic hardship, and the depletion of scarce resources. FAO [10] predicts that 1.3 billion tons of food are lost annually globally each year, costing the economy $1 trillion, the environment $700 billion, and society $900 billion. At once, FAO [5] reported that at least 1.3 billion tons of food were lost or wasted globally in 2007, amounting to roughly one-third of all food produced for human consumption at the time. Food losses and waste deny access to food to the poor in developing countries, lessen resources such as land, water, and fossil fuels, and enhance greenhouse gas emissions associated with producing food [5, 7].

Food loss and waste, according to several studies, are also major contributors to greenhouse gas emissions, which contribute to climate change by creating a greenhouse effect in the earth's atmosphere. Food waste frequently ends up in landfills in developing countries, where it decomposes and emits large amounts of greenhouse gases (GHGs) such as methane and carbon dioxide [11]. Since the Industrial Revolution, anthropogenic greenhouse gases (GHG) such as carbon dioxide (CO2), methane (CH4), and nitrous oxide (N2O) have increased in the atmosphere, causing climate change [12]. Food waste can affect the climate because it produces methane gas as it decomposes in landfills. Methane gas is 25 times more harmful than carbon dioxide as a greenhouse gas. Rice cultivation, as well as increased use of synthetic fertilizers and organic soil cultivation, all contributed to an increase in Methane (CH4) and Nitrous oxide (N2O) emissions in Indonesia [13].

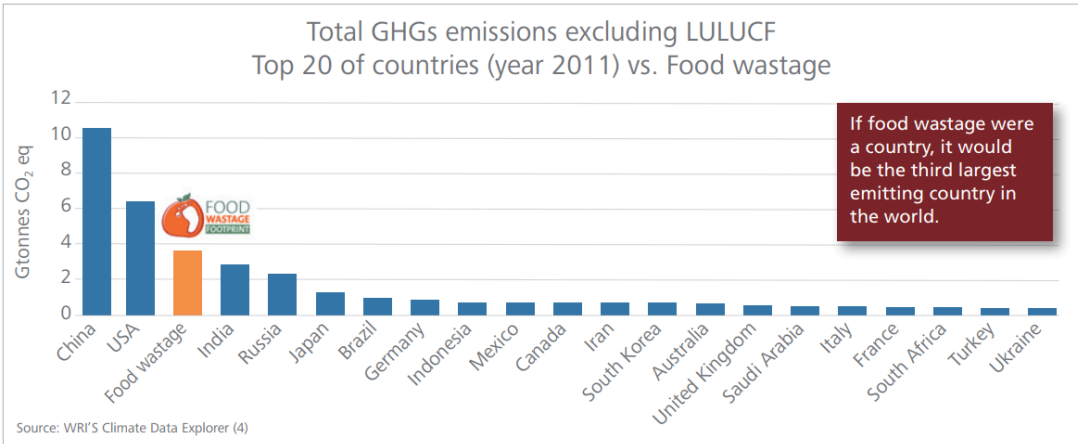

Figure 4. Top 20 of GHG emitting countries vs. food wastage [14]

Besides, the food supply chain accounts for approximately 21-37 percent of the overall GHG emissions. Farming and agricultural use, warehousing, transportation, packaging, manufacturing, wholesaling, and consumption are all examples [2]. According to FAO [5], 8 percent of annual greenhouse gas emissions (about 4.4 gigatons) are formed as a result of producing food which is either lost or wasted. This implies that economic waste emissions contribute nearly as much (87%) as road transport emission levels to global warming [5, 15]. It shows that the value of the emissions is roughly three times that of global aviation emission levels. Or, if we consider national emissions, it would be the third largest emitter in the world. Only China (21%) and the United States (13%) have produced more greenhouse gases (see Figure 4). The implications of food related GHG emissions are expected to rise in the near future as global population and agricultural product consumption rise concurrently [2].

5. Scenario for preventing food and waste in Indonesia

In terms of influence on our way of eating on global warming, the IPCC report on climate change [7] stated that reducing food loss and waste would help to cut carbon emissions and improve food security. If food losses and waste in Indonesia, for example, can be cut in half, the challenge of creating sustainable food and nutritional security will be much easier [11], and the tension to clear forest for new rice fields or agricultural areas will be significantly lowered. Moreover, according to Munesue and Masui [11], 50 percent reductions in food loss and waste are thought to be achievable in the long run, but they are also ideal short-term goals. Furthermore, if the amount of food waste is reduced, the landfill will be smaller, resulting in less methane and greenhouse gas emissions into the atmosphere. This also implies that global warming and climate change will have less of an impact on our planet than previously.

As stated before, generally in Indonesia, food waste occurs at all stages of the food supply chain, including wholesale and retail markets [3]. Some techniques for preventing food losses and waste during the stages of processing, delivery, and consumption stages include revising supermarket standards for high appearance quality for fresh products, selling closer to consumers, developing markets for substandard products, and raising public awareness [16].

5.1. Agricultural production stage

Food may be lost due to premature harvesting for most farmers in developing countries, including Indonesia. Because of food insecurity or a desperate need for cash in the second half of the agricultural season, poor farmers may harvest crops too early. As a result, the food loses nutritional and economic value and may be thrown away if it is unfit for human consumption [16].

To reduce food waste and waste of agricultural products, small farmers can be organized, and their production and marketing can be diversified and scaled up [16]. Small resource-poor farmers can be organized into groups to produce a wide variety of significant amounts of cash animals or plants.

Throughout this case, those who can obtain credit from agricultural financial institutions or advance payments from produce buyers. Moreover, Reduced food losses require the use of appropriate food technology and agricultural machinery at all stages of food handling and product development [17].

5.2. Post-harvest handling and storage

FAO [16] stated that post-harvest agricultural wastes occur in developing countries due to inadequate storage facilities and a poor infrastructure. Fresh fruits, vegetables, meat, and fish can end up ruining in hot temperatures caused by lack of transportation, storage, temperature control, and market infrastructure.

Based on data form APRI [18], Indonesia currently lacks a cold chain for agricultural products. Reportedly, the potential of cold storage facilities is only around 3,000 units with an ability of 15,000 tons per day, or just 10% of needs, that reach an estimate of 150,000 tons per day. This is certainly an unfortunate, given that the agricultural sector, particularly livestock, is a potential market for the cold chain sector. As a result, Indonesia requires facilities and cold chains to support domestic cargo in order to optimize agricultural product transportation. Therefore, the Indonesian government must prioritize infrastructure and transportation investment. Consequently, government should improve road, energy, and market infrastructure including improve storage and cold chain facilities, as well as transportation [19].

5.3. Distribution

In Indonesia, food losses and waste are caused by insufficient market systems and high 'appearance quality standards' from supermarkets for fresh products [3, 16]. Some produce is rejected at the farm gate by supermarkets due to stringent quality standards governing crop weight, size, shape, and appearance. Furthermore, wholesale and retail markets in developing countries including Indonesia are frequently small, overcrowded, unsanitary, and lack cooling equipment [20]. As a result, large portions of crops are never harvested.

Even though the private sector should ideally develop wholesale and retail markets, local governments and marketing cooperatives can help establish and improve market facilities [20]. Furthermore, farmers' commodities must reach consumers in a timely and efficient manner. To that end, selling farm crops closer to consumers without having to meet the strict quality standards set by supermarkets on weight, size, and appearance may reduce the amount of rejected crops.

Despite the fact that some rejected crops are used as animal feed, quality standards may divert food originally intended for human consumption to other uses [3]. As a result, marketing cooperatives are organizations that provide a central point for gathering produce from small farmers and preparing commodities for transportation to markets and other distribution channels in order to reduce food losses and waste during the distribution stage. By increasing the efficiency of these activities, marketing cooperatives should be able to reduce food loss and waste.

5.4. Consumption

Food waste is related to buying and eating behaviour patterns. A campaign to raise awareness of the value of wasted food should be launched to reduce food waste [21]. Food waste occurs in response of issues with mindset and culture, as well as a lack of public awareness about the economic value of food. As a result, attempts to reduce food waste in Indonesia can be carried out through education and awareness that use religious teachings and local wisdom to enhance public awareness and understanding of losing food's economic value in achieving long-term food security.

As acknowledged by Rutten [22], the need for further study to investigate the outcomes of reducing food losses in the supply chain (agriculture, processing, storage, and transport), reducing food loss and waste in the rest of the world, and establishing more comprehensive broader economic impacts throughout household members. This is intended to be Indonesian national policy, and its implementation should involve all stakeholders, including governments (central and local), private

companies (including state-owned enterprises), and communities. Policies, markets, institutions, and governance must all collaborate to develop enabling environment for climate change mitigation and adaptation across the Indonesian food supply chain.

6. Conclusion

Reducing food loss and wastes in developing countries such as Indonesia can help to mitigate hunger while also attempting to make sustainable use of natural resources by cutting greenhouse gas emissions. However, regulations such as social welfare programs to improve access to food in developing countries should be supplemented by solutions to reduce food losses and waste, because poverty, particularly in rural areas, is the underlying cause of food insecurity in developing countries. Consequently, while increased crop yields are necessary, they are insufficient to alleviate hunger and malnutrition, as regularly explained. To reduce the total amount of food losses and waste, intergovernmental solutions especially if it involves changes in consumer purchasing, distribution marketing channels, and public awareness are required.

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